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David Mikhail, Dai Sugimoto, Meena Tadros, Robert Van Pelt, Luke Radel, David Soma, Sophia Ulman, Shane Miller & Jacob Jones

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INJURY DIFFERENCES BETWEEN YOUTH MALE AND FEMALE AMERICAN FOOTBALL PLAYERS

ABSTRACT

Objectives: American football continues to be a popular and rapidly growing sport among both males and females, with relatively little research concerning injuries for the latter – especially in the youth population. The current study aims to evaluate the differences in the epidemiology of injured body parts and injury diagnoses between youth male and female American football players.

Methods: Injury data for male and female American football players were collected over a 10-year period from the National Electronic Injury Surveillance System (NEISS) and retrospectively reviewed. Patients between ages 4 and 18 years with a medical injury/condition related to American football were included. Descriptive analyses were performed for all data including injured body parts, injury diagnoses and setting, demographics, and medical disposition. Odds ratio (OR), 95% CI, and p-values ($p<0.05$) were used for the statistical significance between the two groups.

Results: A total of 82,174 American football injuries were found in the NEISS dataset, 95.0% of which were male with a mean age of 13.0±2.9 years. The most injured body parts were the head (17.7%), finger (13.8%), knee (9.2%), shoulder (8.7%), and ankle (8.0%). The most prevalent injury diagnoses were fractures (24.5%), sprains/strains (22.7%), contusions/bruises (12.2%), concussions (9.8%), and internal injuries (7.1%). Males were more likely to suffer head (OR=1.32, $p<0.001$), knee (OR=1.23, $p<0.001$) and shoulder (OR=2.19, $p<0.001$) injuries and females suffered more finger (OR=3.22, $p<0.001$) and ankle (OR=1.25, $p<0.001$) injuries. Males were more likely to suffer concussions (OR=1.50, $p<0.001$) whereas females sustained more sprains/strains (OR=1.32, $p<0.001$).

Conclusion: Youth male American football players had a greater likelihood of sustaining concussions and head, knee, and shoulder injuries, whereas females were more likely to have sprains/strains, finger, and ankle injuries. Future studies are warranted to further explore how these differences may be attributed to play style and sex-based physiologic and development differences.

INTRODUCTION

American football continues to be a popular and rapidly growing sport among both males and females. According to the National Federation of State High School Associations, participation in traditional tackle football has risen by 5.5% and 5.6% between the 2021-2022 and 2022-2023 school years among males and females, respectively. In addition to tackle football, flag football had 46% and 33% increases in participation among males and females, respectively.[1] Among high schoolers, playing school-affiliated traditional tackle football remains male-dominant, while flag football is gaining more popularity with female athletes.[1, 2]

While the popularity of American football is increasing among females, especially flag football, there is relatively little research concerning this population. To the contrary, previous studies have evaluated patterns of musculoskeletal injuries among male adolescent football players.[3, 4, 5, 6, 7] Some studies have included female football-related injuries, but did not distinguish them from their male counterparts.[8, 9, 10] One study among adults demonstrated that female American football players were more likely to be admitted to the emergency room with a head injury compared to males.[11] Furthermore, although few studies directly compare injuries

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between males and females in football, one study examined 47 youth game-related flag football injuries and found higher injury risk (%) and rate (per 1000 athlete exposures) in females compared to males.[12]

In summary, American football has continued to increase in popularity among males and females through both traditional tackle football and flag football. Although many studies have been conducted comparing the epidemiology of injuries between male and female athletes in other sports such as basketball,[13, 14, 15] baseball/softball,[16] soccer, [17, 18, 19] volleyball,[20] gymnastics,[21, 22] and hockey, [23, 24] comparisons of the injury profiles between males and females in American football, especially among the youth population are limited. Thus, the current study aims to evaluate the differences in the epidemiology of injured body parts and injury diagnoses between youth male and female American football players.

METHODS

Study design and Data Source

A descriptive epidemiological study design was employed for the current study. Injury data related to both male and female American football players were collected from the National Electronic Injury Surveillance System (NEISS) and were retrospectively reviewed. The NEISS, a part of the Consumer Product Safety Commission in the United States (US), represents emergency room visits from over 100 emergency departments across the US.[25] The dataset is a nationally representative sample of more than 5,000 hospitals in the US.[25] A designated staff member at each participating emergency department enters basic demographic information for every patient such as chronological age, biological sex, and racial/ethnic background and medical complaints including affected body parts, medical diagnoses, and event setting.[25] An NEISS coder then reviews the data ensuring that it meets NEISS criteria.[25] The data are then reviewed for their accuracy and appropriateness. Once formally accepted, the patients are added to the NEISS database. An institutional review board approval was obtained prior the commencement of the current study.

Data on male and female American football players over a 10-year period (January 1, 2014 – December 31, 2023) were extracted from the NEISS database. Patients between ages 4 and 18 years with a documented medical injury/condition related to American football were included. Data that did not meet the above criteria were excluded. Additionally, the data that appeared incomplete and/or unclear to interpret were considered invalid so they were excluded. Decisions of data elimination were determined by multiple board-certified sports medicine physicians.

Main Outcome Measures and Data Classifications

Primary outcome variables were injured body parts and injury diagnoses. In addition to the primary outcomes, demographic data such as chronological age and ethnic/racial background were extracted. Furthermore, injury setting and medical disposition information were also extracted from the NEISS database.

Statistical Analysis

Descriptive statistics were initially employed. For continuous variables, mean, standard deviations, and 95% confidence interval (95%CI) were used. For categorical variables, absolute numbers and relative percentages (%) were used. Descriptive analyses were performed for all

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data including injured body parts, injury diagnoses, demographics, injury setting, and medical disposition. For the comparisons, chi-square tests were employed to assess if there were proportional differences in the top five injured body parts and injury diagnoses between male and female American football players. Odds ratio (OR), 95%CI, and *p*-values ($p<0.05$) were used for the statistical significance between the two groups. All analyses were performed by Statistical Package for Social Sciences (IBM SPSS Statistics-V28, Chicago, IL, US).

RESULTS

Demographics

A total of 82,174 injuries related to American football were found in the NEISS dataset: 78,073 (95.0%) in male patients and 4,101 (5.0%) in females. The mean age was 13.0 ± 2.9 years (males: 13.1 ± 2.9; females: 12.7 ± 2.9). The top three racial/ethnic backgrounds consisted of Caucasian/white (35.8%, N=29,456), African American/black (28.9%, N=23,739), and Not Specified (28.8%, N=23,631). Details of the age and racial/ethnic background are presented in Table 1.

Descriptives of Injured Body Parts

The most injured body parts were the head (17.7%, N=14,525), finger (13.8%, N=11,326), knee (9.2%, N=7,548), shoulder (8.7%, N=7,150), and ankle (8.0%, N=6,542). These body part injury details are presented in Table 2. Also, the top five injured body parts are shown separately in Figure 1 (males) and 2 (females).

Descriptives of Injury Diagnoses

The most prevalent injury diagnoses were fractures (24.5%, N=20,155), sprains/strains (22.7%, N=18,619), contusions/bruises (12.2%, N=9,950), concussions (9.8%, N=8,044), and internal injuries (7.1%, N=5,797). A large portion of injuries, 15.9%, were labeled as "other" and not included in any category of injury diagnoses. Specific injury diagnoses are shown in Table 3.

Comparisons between Males and Females: Injured Body Parts

The top five injured body parts (head, finger, knee, shoulder, and ankle) were compared between male and female football players. Males were more likely to suffer head (OR=1.32, 95%CI=1.21, 1.44), knee (OR=1.23, 95%CI=1.10, 1.39), and shoulder (OR=2.19, 95%CI=1.88, 2.55) injuries and females suffered more finger (OR=3.22, 95%CI=3.01, 3.45) and ankle (OR=1.25, 95%CI=1.12, 1.39) injuries. The differences in percentages of each injury between the two groups in each injured body part along with *p*-values are shown in Table 4.

Comparisons between Males and Females: Injury Diagnoses

The top five diagnoses (fractures, sprains/strains, contusions/abrasions, concussions, and internal injuries) were also compared between the two groups. Males were more likely to suffer concussions (OR=1.50, 95%CI=1.32, 1.69) whereas females sustained more sprains/strains (OR=1.32, 95%CI=1.23, 1.42). There were no differences between the two groups in diagnoses of fractures (OR=1.03, 95%CI=0.96, 1.11), contusions/bruises (OR=1.01, 95%CI=0.91, 1.11), or internal injuries (OR=1.13, 95%CI=0.99, 1.27). The details of the analysis including percentages of diagnoses in the two groups with *p*-values are presented in Table 5.

Descriptives of Injury Setting

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The top three most common injury settings were sports (57.7%, N=47,454), unknown (20.5%, N=16,846), and school (17.7%, N=14,534). The top three injury setting ranks were consistent when it was examined males and females separately [males: sports (58.8%, N=45,925), unknown (20.0%, N=15,585), and school (17.3%, N=13,529); females: sports (37.3%, N=1,528), unknown (30.7%, N=1,261), and school (24.3%, N=995)]

Descriptives of Medical Disposition

The top three most frequent disposition recorded were transported & released (95.5%, N=78,508), transported & admitted (2.7%, N=2,255), and left without being seen (1.2%, N=982). The top three disposition patterns were the same when the data for males and females were analyzed independently [males: transported & released (95.4%, N=74,497), transported & admitted (2.8%, N=2,215), and left without being seen (1.2%, N=941); females: transported & released (97.8%, N=4,010), transported & admitted (1.0%, N=40), and left without being seen (1.0%, N=40)].

DISCUSSION

Our study showed that males suffered more head, knee, and shoulder injuries and were diagnosed with more concussions whereas females suffered more ankle and finger injuries with more diagnoses of sprains and strains.

Our overall data showed that the most commonly injured body parts among all participants, male and females, consisted of head (17.7%), finger (13.8%), knee (9.2%), shoulder (8.7%), and ankle (8.0%). This finding was similar to a study by Kaplan et al., which showed that the most commonly injured body sites among post-high school male and female participants in flag football were the fingers (30%), wrist (17%), head (17%), and ankle (13%).[26] The variation in our findings from the study by Kaplan et al. may reflect differences in the study populations. The participants of their study were exclusively flag football players whereas our study included data from all types of American football including tackle. Thus, seeing a larger proportion of head injuries is expected. The overall similarities in injury locations between two studies may be attributed to the nature of the sport: catching balls traveling at high speeds, collisions with other players, falls, etc. Previous studies have shown that head injuries are usually among the most common injuries in young football players.[12, 27, 28, 29] Regarding finger injuries, every style of (traditional tackle football, flag football, or two-hand touch) requires players to catch a ball traveling at high speeds, while the players are actively running, placing the fingers in a vulnerable position for a variety of injuries. Additionally, finger injuries may occur when a player grabs an opponent's jersey, pulls flags in flag football, or tackles or blocks opponents. These maneuvers all predispose athletes to a multitude of injury types, such as fractures of the phalanges or metacarpals, finger dislocations, mallet finger, or jersey finger.[30, 31] Larger joints such as the ankle and knee are common areas for injury in sports that require rapid changes in direction and players falling on each other, resulting in atypical stresses on the joints.[32, 33, 34]

Among our composite population, the most common injury diagnoses were fractures (24%), sprains/strains (22.7%), contusions/bruises (12.2%), concussions (9.8%), and internal injuries (7.1%). According to a study by Shankar et al. among male high school football players, the most common injury types were sprains/strains (35.5%) followed by fractures (28.4%) and contusions and bruises (14.3%).[6] The differences in the top diagnosis may be due to different data sources. The current study data were obtained from the NEISS, which typically includes injuries serious enough to warrant urgent care and/or emergency room access. Thus, severe

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injuries such as fractures may emerge as a top diagnosis compared to the sprains/strains seen using a high school injury surveillance system as that used by Shankar et al. In addition, the mean age of the current study (Table 1) is 13 years old compared to the high schoolers considered in the study by Shankar et al. Differences in skeletal maturity may also be contributing to these differences. Nevertheless, the patients considered in both studies were pediatric patients who may still have some degree of skeletal maturation remaining. In combination with fracture susceptibility present at physes, fractures are a common occurrence among young athletes.[36]

The differences in injured body parts between male and female participants may be attributed to a multitude of factors. Our data indicated that head injuries were more likely to be suffered by male participants, while female participants had a greater likelihood of suffering finger injuries. We initially hypothesized that males would have higher rates on head injuries due to a greater involvement in tackle football.[1] While this was in line with our data, the higher likelihood of head injuries in males in our study differed from what was seen by Foss et al.; however, they examined game-related injuries exclusively.[12] Another study by Peterson et al. found that the rate for concussions per exposure in tackle football was 0.51 times that of flag football.[29] Nevertheless, the hypothesis that participants sustain more head-related injuries as the degree of contact increases has been validated by data from other studies such as that by Waltzman et al. which demonstrated that among young male participants, there are significantly more head injuries in tackle football compared to flag football, though players of both types of football experience considerable amounts of head injuries.[37] Furthermore, similarly to our study, Collins et al. found that among women participating in intramural American flag football, the most common sites of injury were the finger (39%), knee (16%), and ankle (8%).[38] Kaplan et al. also noted a relatively higher number of finger injuries among flag football athletes.[26, 39] The higher rates of finger injuries may thus be attributed to more participation in flag football since females participate in flag football at much higher rates compared to tackle.[1]

When comparing injury diagnoses between sexes, males were more likely to undergo concussions, while female participants had a greater likelihood of suffering strains/sprains. It is known that females reach skeletal maturity more quickly compared to males.[40] As a result, their limbs are closer to their final adult size, resulting in relatively heavier limbs that require more force to move them and increased muscle tightness, indicating they may be at greater risk for sprains/strains.[34, 41] Previous studies have demonstrated that females are at higher risk for ankle sprains and ligamentous injuries.[42, 43] This may be due to hormonal influences on musculoskeletal tissue and differences in motion biomechanics in females.[44, 45] Our data supports this, as female participants did experience a higher likelihood of sprains/strains compared to males.

Limitations

There are a few limitations in the current study. First, female American football players comprised only 5% compared to 95% of the data from males. This was expected given that females make up a small minority of all football participants. Second, the NEISS data includes only musculoskeletal injuries that presented to the emergency department. Thus, injuries that were treated by athletic trainers or other medical professionals on the sideline, at home, or in the clinical setting were not included. Third, the context of the injury was not considered such as recreational football versus formal competitive play, or the style of football being played. As a result, we could not differentiate injuries recorded in the NEISS based on their context. Fourth,

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the data collected relied on emergency department medical providers' documentation so potential recall bias could not be eliminated in injury reporting. Finally, data regarding ongoing clinical status and final injury outcome are not included in the NEISS database. As a result, all injuries, regardless of severity or long-term outcomes, were treated as it was recorded in the NEISS database.

CONCLUSION

Youth male American football players had a greater likelihood of sustaining head, knee, and shoulder injuries, whereas females were more likely to have finger and ankle injuries. Regarding diagnoses, males were more likely to suffer concussions, and females had a greater likelihood of sprains/strains. These differences may be attributed to differences in tackle and flag participation between each sex and physiological differences between males and females. Future studies are warranted to further explore how these differences may be attributed to play style, physiologic differences between males and females, and differences attributed to development.

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Table 1. Age and Race/Ethnicity of the Male and Female American Football players.

	Male	Female	All
Age (years)	13.1 ± 2.9 (13.0, 13.1)	12.7 ± 2.9 (12.6, 12.8)	13.0 ± 2.9 (13.0, 13.1)
Race/Ethnicity			
Caucasian/White	27,913 (35.8%)	1,543 (37.6%)	29,456 (35.8%)
African American/Black	22,723 (29.1%)	1,016 (24.8%)	23,739 (28.9%)
Not specified	22,433 (28.7%)	1,197 (29.2%)	23,631 (28.8%)
Others	4,125 (5.3%)	289 (7.0%)	4,414 (5.4%)
Asian	459 (0.6%)	35 (0.9%)	494 (0.6%)
American Indian/ Alaska Native	229 (0.3%)	14 (0.3%)	243 (0.3%)
Not Hispanic/ Pacific Islander	191 (0.2%)	7 (0.2%)	198 (0.2%)

The age was expressed as mean ± standard deviation, 95% CI values.

Table 2. Absolute and Relative (%) Numbers of Injured Body Parts.

	Male	Female	All
Head/Trunk	**17,267 (22.2%)**	**792 (19.3%)**	**18,059 (22.0%)**

Head	13,944 (17.9%)	581 (14.2%)	14,525 (17.7%)
Non-specific facial	2,125 (2.7%)	129 (3.1%)	2,254 (2.7%)
Eye-related injuries	359 (0.5%)	39 (1.0%)	398 (0.5%)
Ear-related injuries	145 (0.2%)	9 (0.2%)	154 (0.2%)
Mouth-related injuries	694 (0.9%)	34 (0.8%)	728 (0.9%)
Neck/Trunk/Back/Spine	**7,988 (10.2%)**	**262 (6.5%)**	**8,250 (10.0%)**
Neck/Cervical spine	1,826 (2.3%)	52 (1.3%)	1,878 (2.3%)
Upper back/Thoracic spine	3,107 (4.0%)	117 (2.9%)	3,224 (3.9%)
Lower back/Lumbar spine	3,055 (3.9%)	93 (2.3%)	3,148 (3.8%)
Upper Extremity	**31,299 (39.3%)**	**2,059 (50.2%)**	**33,358 (40.6%)**
Shoulder/Clavicle	6,974 (8.9%)	176 (4.3%)	7,150 (8.7%)
Upper arm	773 (1.0%)	22 (0.5%)	795 (1.0%)
Elbow	2,213 (2.8%)	86 (2.1%)	2,299 (2.8%)
Forearm	4,361 (5.6%)	134 (3.3%)	4,495 (5.5%)
Wrist	4,341 (5.6%)	205 (5.0%)	4,546 (5.5%)
Hand	2,629 (3.4%)	118 (2.9%)	2,747 (3.3%)
Finger	10,008 (12.8%)	1,318 (32.1%)	11,326 (13.8%)
Lower Extremity	**20,435 (26.3%)**	**958 (23.5%)**	**21,393 (26.1%)**
Thigh	1,059 (1.4%)	24 (0.6%)	1,083 (1.3%)
Knee	7,233 (9.3%)	314 (7.7%)	7,547 (9.2%)
Lower leg	3,256 (4.2%)	75 (1.8%)	3,331 (4.1%)
Ankle	6,146 (7.9%)	396 (9.7%)	6,542 (8.0%)
Foot	2,042 (2.6%)	117 (2.9%)	2,159 (2.6%)

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Toe	699 (0.9%)	32 (0.8%)	731 (0.9%)
Other/Unspecified	1,084 (2.0%)	30 (0.5%)	1,114 (1.3%)
Total	**78,073 (100.0%)**	**4,101 (100.0%)**	**82,174 (100.0%)**

Table 3. Absolute and Relative (%) Numbers of Injury Diagnoses.

	Male	Female	All
Bony/Joint related	**21,590 (27.7%)**	**1,122 (27.3%)**	**22,712 (31.3%)**
Avulsion	61 (0.1%)	9 (0.2%)	70 (0.1%)
Fracture (traumatic)	19,129 (24.5%)	1,026 (25.0%)	20,155 (24.5%)
Subluxation/Dislocation	2,400 (3.1%)	87 (2.1%)	2,487 (3.0%)
Muscle/Tendon/Ligamentous	**17,487 (22.4%)**	**1,132 (27.6%)**	**18,619 (35.5%)**
Strain/Sprain	17,487 (22.4%)	1,132 (27.6%)	18,619 (22.7%)
Other Soft Tissue related	**12,533 (16.0%)**	**620 (15.1%)**	**13,154 (17.4%)**
Contusion/Bruise	9,763 (12.5%)	505 (12.3%)	10,268 (12.5%)
Laceration/Abrasion/Other Skin-related injuries	2,770 (3.5%)	115 (2.8%)	2,886 (3.5%)
Others	**26,465 (33.9%)**	**1,227 (30.0%)**	**27,689 (15.8%)**
Concussion	7,762 (9.9%)	282 (6.9%)	8,044 (9.8%)
Dental injury	222 (0.3%)	5 (0.1%)	227 (0.3%)
Internal injury	5,537 (7.1%)	260 (6.3%)	5,797 (7.1%)
Miscellaneous	12,944 (16.6%)	680 (16.7%)	13,621 (16.3%)
Total	**78,073 (100.0%)**	**4,101 (100.0%)**	**82,174 (100.0%)**

Table 4. Comparisons of Injured Body Parts between Male and Female American Football Players

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Injured Body Part	Males	Females	P-value	OR with 95%CI
Head	17.9%	14.1%	**<0.001**	1.32 (1.21, 1.44)
Finger	12.8%	32.1%	**<0.001**	3.22 (3.01, 3.45)
Knee	9.3%	7.7%	**<0.001**	1.23 (1.10, 1.39)
Shoulder	8.9%	4.3%	**<0.001**	2.19 (1.88, 2.55)
Ankle	7.9%	9.7%	**<0.001**	1.25 (1.12, 1.39)

The percents (%) were recorded injuries by each group.
The OR with 95%CI was calculated based on a group with low percents (%).

Table 5. Comparisons of Injury Diagnoses between Male and Female American Football Players

Injury Diagnosis	Males	Females	P-value	OR with 95%CI
Fracture	24.5%	25.0%	0.453	1.03 (0.96, 1.11)
Sprain/Strain	22.4%	27.6%	**<0.001**	1.32 (1.23, 1.42)
Contusions/Bruise	12.1%	12.0%	0.900	1.01 (0.91, 1.11)
Concussion	9.9%	6.9%	**<0.001**	1.50 (1.32, 1.69)
Internal Injury	7.1%	6.3%	0.067	1.13 (0.99, 1.27)

The percents (%) were recorded injuries by each group.
The OR with 95%CI was calculated based on a group with low percents (%).

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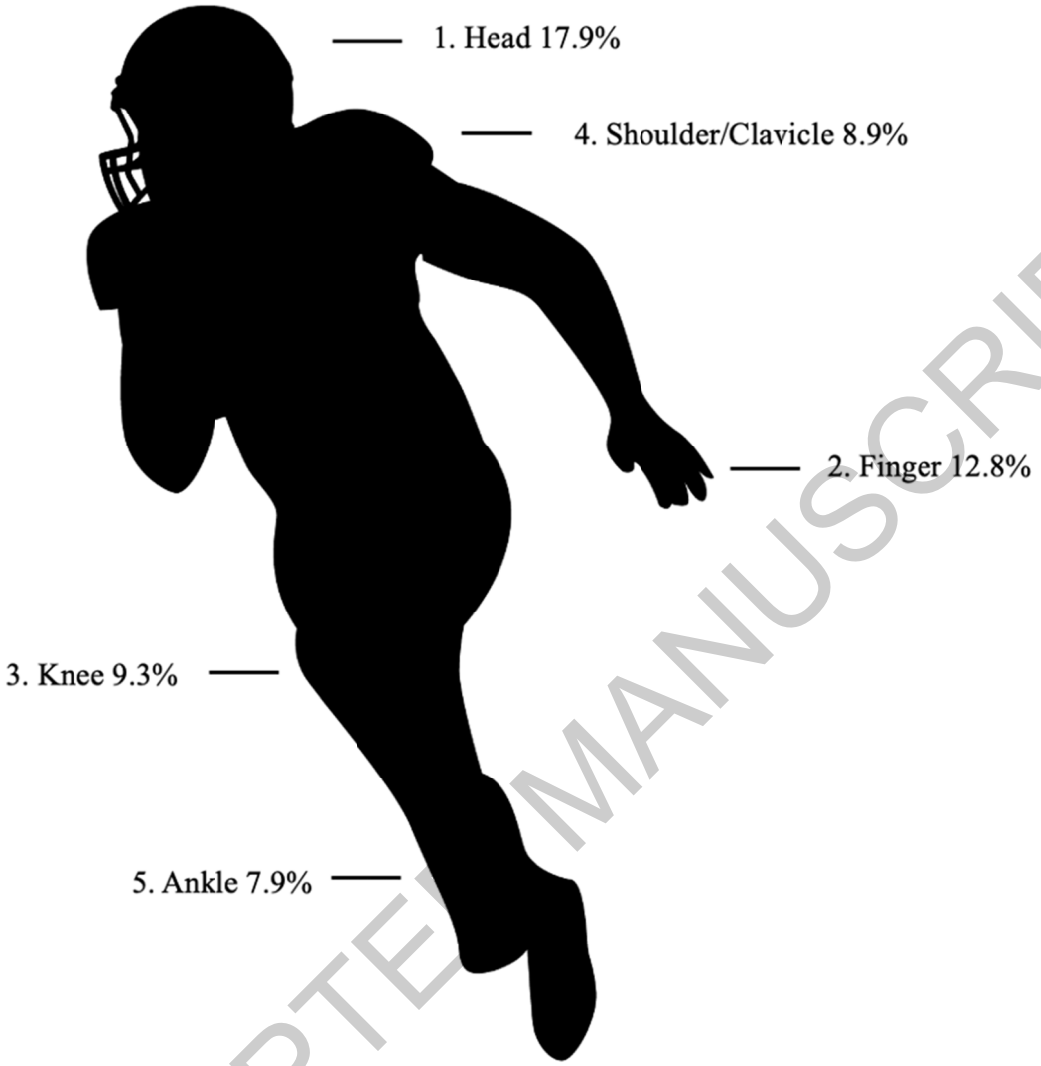

Figure 1. The Top Five Injured Body Parts in Youth Male American Football Players

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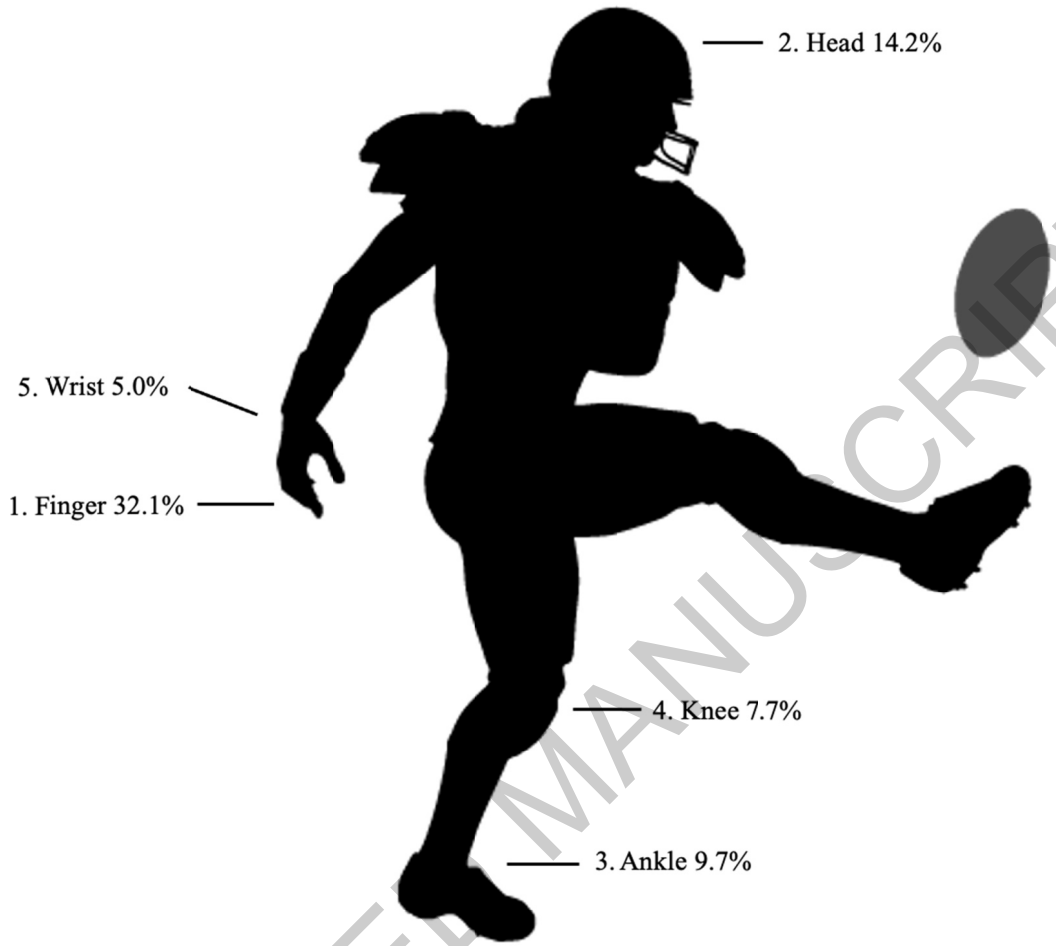

Figure 2. The Top Five Injured Body Parts in Youth Female American Football Players

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INJURY DIFFERENCES BETWEEN YOUTH MALE AND FEMALE AMERICAN FOOTBALL PLAYERS

David Mikhail, BS[1], Dai Sugimoto, PhD, ATC[2, 3], Meena Tadros, BS[1], Robert Van Pelt, MPH[4], Luke Radel, MD[5], David Soma, MD[5], Sophia Ulman, PhD[1, 4], Shane Miller, MD[1, 4], Jacob Jones, MD[1, 4]

1. University Texas Southwestern Medical Center, Dallas, TX, USA
2. Waseda University, Tokyo, Japan
3. The Micheli Center for Sports Injury Prevention, Waltham, MA, USA
4. Scottish Rite for Children, Dallas, TX, USA
5. Mayo Clinic, Rochester, MN, USA

Corresponding Author: David Mikhail, BS

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